

REAL ESTATE PRICE PREDICTION AND INVESTMENT ANALYZER

AI-Powered Property Intelligence Platform

MAJOR PROJECT REPORT

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Programme: Bachelor of Computer Applications (BCA)

University: Silver Oak University

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Certificate

This is to certify that the major project titled **Real Estate Price Prediction and Investment Analyzer** has been prepared as an academic project in partial fulfillment of the requirements of the Bachelor of Computer Applications programme. The work demonstrates the application of Python, machine learning, data analytics and Streamlit-based dashboard development.

Project Guide: _____ HOD: _____ Date: _____

Declaration

I declare that this project report and the accompanying software have been prepared for academic purposes. The included demonstration dataset is synthetic and is not claimed to represent verified real-estate transactions. Any future production use must rely on appropriately licensed, documented and validated data.

Acknowledgement

I express my sincere gratitude to my project guide, faculty members, university and peers for their guidance and support during the development of this project. I also acknowledge the open-source Python ecosystem, including Pandas, NumPy, scikit-learn, Streamlit and Folium, which provides the technical foundation for this academic prototype.

Abstract

Real Estate Price Prediction and Investment Analyzer is an AI/ML-based academic decision-support platform designed to combine property valuation, locality intelligence and investment analysis in one interactive dashboard. The system processes property-level attributes such as area, bedrooms, bathrooms, age, furnishing, parking, proximity to metro stations, schools and hospitals, crime index, historical price growth and future infrastructure score. Two regression approaches—Gradient Boosting and Random Forest—are trained and evaluated using MAE, RMSE and R^2 . The best-performing model is packaged for interactive predictions through a Streamlit interface. The platform also provides locality summaries, price-per-square-foot analysis, rental yield, loan EMI, projected future value, ROI estimates, property comparison, a transparent deal scorer and map-based area exploration with heatmap and circle-drawing tools. The project is intentionally supplied with synthetic demonstration data so that the complete software can be executed without making unsupported claims about real market records. The architecture is designed so legitimate real-estate datasets can later replace the demo CSV files without changing the core application flow. The result is a practical major-project prototype showing how machine learning and geospatial analytics can support structured property research.

Keywords: Real Estate, Machine Learning, Gradient Boosting, Random Forest, Streamlit, Investment Analysis, Rental Yield, Geospatial Analytics

1. Introduction

Real estate decisions involve many interacting factors: property characteristics, locality demand, access to transport and amenities, safety, future infrastructure and historical market movement. A user may therefore need several disconnected tools to estimate value and assess an investment. This project integrates those tasks into one academic platform.

2. Problem Statement

The project addresses the problem of estimating property value from heterogeneous features and converting the estimate into understandable locality and investment indicators. The system should provide repeatable calculations rather than relying only on subjective judgement.

3. Objectives

- Build a property-price prediction pipeline.
- Compare machine-learning regressors.
- Analyze locality-level factors.
- Calculate rental yield, EMI, future value and ROI.
- Score a listing against predicted market value.
- Provide an interactive map and area analysis.
- Package the result as a GitHub-ready academic project.

4. Scope

The prototype focuses on residential property intelligence and Ahmedabad-based synthetic demonstration records. It can be adapted to other cities by replacing the data and retraining the models. The scope includes valuation, analytics, investment calculations, comparison and map visualization; it does not provide legal, brokerage or financial guarantees.

5. Existing System

Conventional property research often uses listing portals, spreadsheets and manual calculations separately. Such workflows can make it difficult to compare properties consistently and may omit locality-level signals.

6. Proposed System

The proposed platform combines a structured data layer, preprocessing pipeline, regression models, prediction utilities, investment formulas, deal scoring and a Streamlit dashboard. The architecture is modular so each component can be tested independently.

7. Functional Requirements

The system shall load property and locality data; validate required inputs; train regression models; predict price; show evaluation metrics; calculate investment indicators; score deals; compare properties; visualize geographic records; and export/extend results through standard CSV-based data structures.

8. Non-Functional Requirements

Usability, reproducibility, maintainability, reasonable response time, clear documentation, modular code, transparent assumptions and safe interpretation of model outputs are key requirements.

9. Technology Stack

Python 3; Pandas and NumPy for data processing; scikit-learn for machine learning; Streamlit for the interactive web application; Folium and streamlit-folium for maps; Matplotlib for documentation diagrams; Joblib for model persistence; ReportLab for this report.

10. Dataset Design

The project includes three CSV datasets: property_transactions.csv, locality_data.csv and infrastructure.csv. The transaction file contains 3,500 synthetic records with property, location, accessibility, safety, infrastructure and pricing variables. The locality and infrastructure files support aggregation and future expansion.

11. Data Preprocessing

Categorical features are one-hot encoded and missing values are handled using imputers. Numerical variables are passed through a preprocessing pipeline. The training target is price_inr. A reproducible random seed is used for train/test splitting.

12. Feature Engineering

The design includes amenity access, development-adjusted growth and price-per-square-foot concepts. The production model currently uses a controlled feature set to keep the prediction pipeline stable and interpretable.

13. Machine Learning Algorithms

Gradient Boosting builds an ensemble of sequential decision trees that reduce prediction error by learning from previous residuals. Random Forest averages many decision trees and can model non-linear relationships. Both models are evaluated on an unseen test split.

14. Model Evaluation

Evaluation uses Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . MAE expresses average absolute prediction error in currency units; RMSE penalizes larger errors more strongly; R^2 measures the proportion of variance explained by the model. The included metrics are valid only for the supplied synthetic dataset.

15. Price Prediction Module

Users enter locality, property type, furnishing, size, room counts, building age, floor information, parking, accessibility distances, crime index, infrastructure score and historical growth. The selected trained model returns an estimated market value and derived price per square foot.

16. Locality Intelligence Module

The module summarizes median property price, price per square foot, rent, growth, infrastructure and safety-related indicators. It also provides a yearly median-price view to help users understand the synthetic dataset's trend.

17. Investment Analyzer

The investment module calculates down payment, loan amount, EMI, gross rental yield, net rental yield, future value, capital appreciation, initial transaction-adjusted investment and an estimated ROI. These are mathematical projections based on user assumptions, not financial advice.

18. Deal Scorer

The deal scorer compares a listing price with the model-estimated value and adjusts a transparent score using locality growth and infrastructure indicators. Labels are Undervalued, Fairly Priced or Overpriced based on the model gap thresholds.

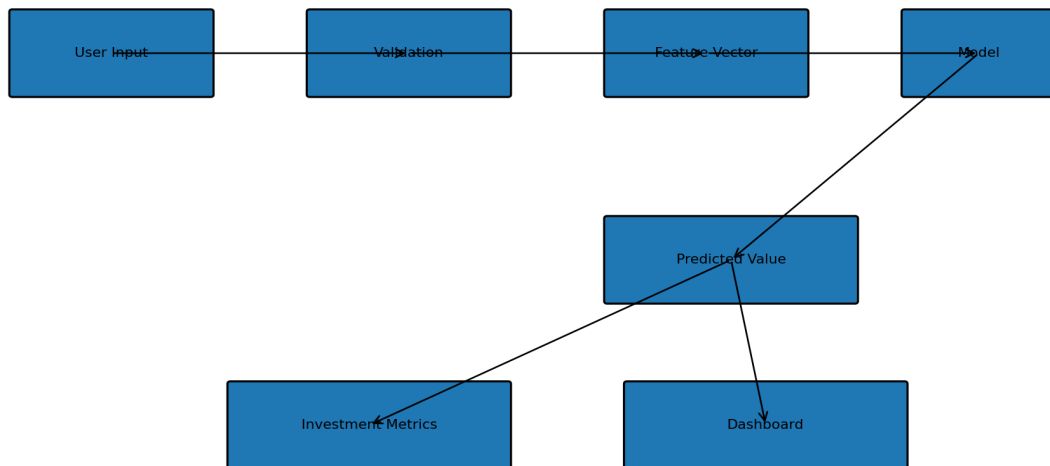
19. Property Comparison

Users can select multiple demonstration properties and compare price, size, rent, growth, infrastructure and price-per-square-foot. This supports structured side-by-side review.

20. Map Intelligence

The map displays property markers and a price heatmap. A Folium drawing tool is provided for circle selection, while manual center and radius inputs provide a reliable area-analysis fallback. Haversine distance is used to identify records within the selected radius.

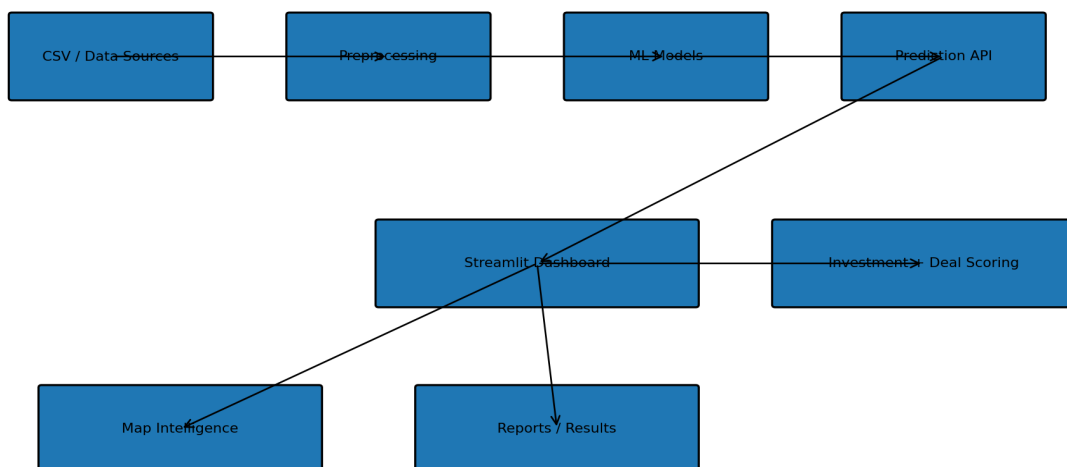
Data Flow



21. System Architecture

The platform follows a layered architecture: data → preprocessing → model → prediction utilities → Streamlit interface. Business calculations and map analysis remain separated from the user interface.

System Architecture



22. Testing

Unit tests cover EMI/future value, deal scoring and positive model prediction. The model training script is executed before the dashboard so the persisted model bundle is available.

23. Results

The supplied demo package trains two models, selects the best model by R^2 , stores model metrics in JSON and persists the winning pipeline with Joblib. Exact metrics are generated during package creation and displayed in the Model Evaluation page.

24. Security, Privacy and Ethics

A production implementation should validate uploaded data, avoid unnecessary personal information, secure any external API keys, document data licenses and avoid presenting model estimates as guaranteed property values. Synthetic data is used here specifically to reduce unsupported claims about individuals or transactions.

25. Limitations

The demonstration data is synthetic; real market dynamics, taxes, legal title, neighborhood microstructure, demand shocks and macroeconomic conditions are not fully represented. The investment formulas also omit some advanced cash-flow, tax and financing effects.

26. Future Scope

Future versions can integrate licensed transaction feeds, real infrastructure GIS layers, automated geocoding, richer time-series forecasting, model monitoring, confidence intervals, user accounts, saved portfolios, scenario analysis and explainability methods such as SHAP where appropriate.

27. Conclusion

The project demonstrates an end-to-end major-project workflow from data design and machine learning to an interactive analytics dashboard. It combines predictive modeling, investment mathematics and geospatial exploration into a single reusable platform.

28. Model Metrics from Included Demo Dataset

Model	MAE (■)	RMSE (■)	R ²
gradient_boosting	813,958	1,147,321	0.9779
random_forest	846,581	1,197,004	0.9760

Selected model: **gradient_boosting**. These results are benchmark results on synthetic demonstration data only.

29. Project Folder Structure

Real_Estate_AI_Major_Project/

- app.py
- requirements.txt
- README.md
- data/
- models/
- src/
- notebooks/
- tests/
- docs/
- assets/
- report/

30. How to Run

```
python -m venv .venv
pip install -r requirements.txt
python src/train_models.py
streamlit run app.py
```

31. References

1. Streamlit Documentation — official guides for building and running data applications.
2. scikit-learn Documentation — regression estimators and model evaluation.
3. Pandas Documentation — tabular data manipulation.
4. Folium Documentation — interactive Leaflet maps from Python.
5. NumPy Documentation — numerical computing.

32 Appendix A — Submission Checklist

Project title: Real Estate Price Prediction and Investment Analyzer

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GitHub: <https://github.com/parthxixu-svg/Real-Estate-AI-Major-Project>

Colab: <https://colab.research.google.com/drive/142YYuV41AQLJyQILgejHhno2iG1GISFw>

Source code: included

Dataset: included (synthetic demo)

Model files: included

Testing: included

Screenshots: to be captured from the running dashboard before final submission.

Academic Disclaimer

This package is an educational prototype. Replace synthetic data with legitimate, documented data before claiming real-world accuracy or using the system for actual property or financial decisions.